

Build your own Decentralised Large Scale Key-Value Cloud Storage



In its simplest avatar, a key-value database stores data as a collection of key-value pairs, with the key serving as the unique identifier for a value. Both the keys and values could be anything from simple objects to complex objects. This article centres around building a large scale key-value storage in the cloud.

A key-value store is a type of NoSQL database where each item is a combination of keys and values. A value can be stored and retrieved by referencing its key. Key-value stores are great for use cases where large amounts of data need to be stored and no complex queries have to be performed during retrieval of data. Examples include storing multimedia, cache management, real-time recommendations and advertising, hierarchical data, event log processing, etc.

Relational databases are not suitable for such use cases as they have limitations like static schema, expensive vertical scaling which is also possible up to a certain limit, only read scalability, single point of failure, and inability to consume data in large volumes. A key-value store helps to overcome these limitations.

This article presents the design and implementation of a scalable, distributed, key-value store based on the design of Pastry (<http://rowstron.azurewebsites.net/PAST/pastry.pdf>) and Tapestry. The application design

is completely decentralised and self-adaptable to the failure of nodes in the application's network, the exit of existing nodes and arrival of new nodes. A message is consistently routed to the relevant live nodes in the application's network, while all nodes are connected through the Internet.

The application's design, which is proposed in this article, can be used as a base for building various peer-to-peer (P2P) applications. Routing and location of messages is performed at the application level. The 'Design Consistent Hashing and CAP Theorem'

